MD. AMANULLAH ASIF

Mechanical Engineer · CSWP · Co-Inventor, U.S. Provisional Patent
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Patent No. 63/935,783 (US) | LinkedIN |Portfolio Website

RESEARCH INTERESTS

- Computational Mechanics (MATLAB/PYTHON/FEM/FDM)
- Electro-Thermo-Mechanical (Multiphysics Coupling) Modeling
- Scientific Machine Learning (PINNs, data-driven mechanics)
- Machine Learning for Alloy Design (HEA/MPEA)
- Composite Structural Integrity
- Multiphysics Simulation
- DFT, AIMD, Spin Quantum Electronics (Valleytronics)

EDUCATION

B.Sc. in **Mechanical Engineering** *Jan 2022 – Jun 2026*

Bangladesh University of Engineering and Technology (BUET), Dhaka

- **CGPA: 3.58 / 4.00 (4th year- 3.79/4.00)**

RESEARCH EXPERIENCE

- **Coupled Structural–Materials Analysis (Undergraduate Thesis)** *Jan 2025 – Present*
  BUET, Dept. of Mechanical Engineering · Supervisor: ***Prof. Dr. Shaikh Reaz Ahmed***
- **AhmedLab- [Undergraduate Researcher]** *Jan 2024 – Present*
  Buffalo State University · Supervisor: ***Prof. Dr. Saquib Ahmed***

Publications

- Mechanistic and Data-Driven Modelling of Mechanical Properties in High-Entropy Alloys. **Amanullah Asif**[a†], Saquib Ahmed[e*] et al. (**First author**)- manuscript **under review** in **Journal of Alloys and Compounds**. [**Pre-print**]
- Spin-Polarized DFT+U Study of Mn-Doped Anatase TiO_2 for Photocatalysis **Amanullah Asif**[a†], Saquib Ahmed[e*] et al. (**Co-first autor**)-manuscript **under review** in **Materials Today Communications. [View Manuscript]**
- Reliability Assessment of Machine Learning Models for Single-Atom Alloy Catalyst Screening: A Chemistry-Aware Validation Framework for Methane C–H Activation Barrier Prediction. Md. Atik Faisal, **Md. Amanullah Asif** et al, (**Co-first author**)- manuscript **Published** in **Computational Materials Today**. [View Manuscript]
- Physics-Constrained Machine Learning Framework for Mechanical Property Prediction in CNT–Al Composites. **Amanullah Asif,** Saquib Ahmed et al.- manuscript **under review** in **Journal**. [Pre-print]

PATENT & INVENTION

U.S. Provisional Patent *Real-Time Image Analytics Platform for Rolling Mill Stability Management*
Patent No. 63/935,783 · Filed: December 2025 · Co-inventors: S. Ahmed, M. Jawad, A. Rumman, A. Faisal, **A. Asif [Available for Licensing at State University of New York]**

KEY ENGINEERING PROJECTS

Coupled Structural–Materials Analysis (Undergraduate Thesis) *Jan 2025 – Present*

BUET, Dept. of Mechanical Engineering · Supervisor: Prof. Dr. Shaikh Reaz Ahmed

- Developing a novel MATLAB-based framework for coupled electro-thermo-mechanical analysis of curved structures using displacement-function and finite-difference methods.
- Investigating the effects of material nonlinearity, boundary constraints, and geometric parameters on structural response through analytical and numerical modelling.
- Predicting elastic parameters through rigorous mathematical modelling and independently validating results using ANSYS FEA; **manuscript under development**. [See available codes and reports]

Valleytronics of WTe2/VSe2 Heterostructure *Jan 2024 – Present*

Associate Professor, SUNY Buffalo State university, USA; Supervisor: Dr. Saquib Ahmed,

- Modeled WTe2/VSe2 heterostructures with SOC and DFT+U. Performed band structure and valley polarization analysis.
- Investigating magnetic proximity effects for future valleytronic devices. Current Status: Optical Conductivity &Valley Valve Filter feasibility optimization ongoing in VASP & Wannier. **Manuscript Preparation Ongoing.** [See Experimental Outcomes]

4 DOF Chess-Playing Robotic Arm *--BUET Undergraduate Electro-Mechanical System Design*

- Developed the complete system through SOLIDWORKS CAD design, mechanical and electronics integration, servo/microcontroller control, motion-planning programming, and real-time validation.
- Designed, fabricated, and tested a fully functional chess-playing robotic arm, taking the system from initial concept to a working prototype. [See the prototype and its functions]

Waste-Heat Recovery Heat Exchanger *--BUET Heat Exchanger Design & Fabrication Project*

- Designed and fabricated a waste-heat recovery heat exchanger targeting industrial thermal efficiency improvement.
- Supported design with thermal and flow simulations (SOLIDWORKS Flow Simulation, ANSYS Fluent/Thermal); validated predictions against experimental data from fabricated prototype. [See the prototype and its functions]

TECHNICAL SKILLS

Programming & Data: Python, MATLAB, Machine Learning (XGBoost, LightGBM, CatBoost, SHAP), Neural Networks, Feature Engineering.
Simulation & FEA: ANSYS (Mechanical, Fluent, Thermal), SOLIDWORKS Simulation, COMSOL, Flow Simulation.
Design & Drafting: SOLIDWORKS (CSWP), AutoCAD, CAD Modelling, GD&T, Technical Drafting
Fabrication & Hardware: Prototyping, Machining, Electronics Integration, Robotics (servo, microcontroller)
Numerical Methods: FEM, FDM, Displacement Function Method, Method of Manufactured Solutions (MMS)

CERTIFICATIONS

1. SOLIDWORKS Certified Professional (CSWP) — Dassault Systèmes. **2.** SOLIDWORKS Certified Associate (CSWA) — Dassault Systèmes. **3.** COMSOL Multiphysics Specialization — 2D Simulations.

INDUSTRIAL TRAINING

Rural Power Company Limited (RPCL) *Completed*

- Hands-on industrial training covering power plant operations and engineering practices.

AWARDS & ACHIEVEMENTS

- University Technical Scholarship — awarded multiple terms for outstanding academic performance at BUET.
- 1st Place, Rajshahi Division — Dakhil (SSC-equivalent) Examination; ranked 1st in Rajshahi Division nationally.
- Aspire Leadership Program — completed advanced leadership and professional development curriculum.